

ORIGINAL ARTICLE

Electroacupuncture on PC6 Prevents Opioid-Induced Nausea and Vomiting after Laparoscopic Surgery*

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ABSTRACT **Objective:** To investigate the treatment time dependence of electroacupuncture (EA) on Neiguan (PC6) for preventing postoperative nausea and vomiting (PONV). **Methods:** One hundred and seventy-eight patients, who had received intravenous patient-controlled analgesia (PCA) with Fentanyl, were assigned randomly to three groups using random numbers: a pre-operative EA group (PrEA), a post-operative EA group (PoEA), and a non-acupuncture control group (NC). An anesthetist evaluated the incidence and severity of nausea and vomiting for 48 h after surgery blindly. The main outcomes were severity and frequency of PONV, which were measured with a self-reported questionnaire and a confirmation from the anesthetist. The data were analyzed with ANOVA and Z-test. **Results:** The incidence of nausea and vomiting was significantly lower in the PrEA group than the NC group during 48 h after surgery ($P<0.01$, $P<0.05$). The incidence of vomiting was also significantly lower in the PrEA group than the PoEA group ($P<0.05$). The PoEA subjects evidenced no significant differences compared with the NC subjects in terms of the incidence of nausea and vomiting ($P<0.05$). The severity of nausea was significantly lower in the PrEA group than in the NC and PoEA groups ($P<0.05$). **Conclusions:** EA on PC6 is effective in the prevention of PONV, and pre-operative acupuncture is more effective than post-operative acupuncture.

KEYWORDS electroacupuncture, Neiguan (PC6), postoperative nausea and vomiting

Postoperative nausea and vomiting (PONV) is a common adverse reaction that delays the discharge date of healthy patients, and frequently causes patients to be re-admitted suddenly.⁽¹⁾ Risk factors for PONV include the patients' age, sex, past history of PONV, smoke, total length of surgery, and opioid medication was employed.⁽²⁾ Patient-controlled analgesia (PCA) is becoming increasingly broadly used for the control of post-operative pain. However, it has also been claimed that opioid medication may be highly relevant to the manifestation of PONV.⁽³⁾

Many types of interventions, including antiemetics or the use of anesthetics known not to cause PONV are often used to prevent PONV, however, these methods are quite expensive and can also cause other adverse reactions. This has compelled patients, practitioners, and researchers to look toward non-pharmacological interventions, including acupuncture-type treatments, that have previously been shown to be effective.⁽⁴⁻⁹⁾

The Neiguan (PC6) has been used widely for the prevention of PONV because patients can easily stimulate it by themselves, even while wearing an apparatus.⁽¹⁰⁾ At the National Institutes of Health (NIH) Consensus Conference in November 1997,

acupuncture was reported to be effective for the treatment of PONV. It was also announced that the time had come for in-depth studies into its effects.⁽¹¹⁾ A systematic review of the effects of PC6 was conducted to determine its PONV-ameliorating effects.⁽¹²⁾ However, it was also acknowledged that further studies were warranted to confirm its effects, depending on the time at which acupuncture was applied. We previously evaluated the anti-nausea effects of electroacupuncture (EA) at PC6 on laparoscopic surgery patients who were in the recovery room; however, the results revealed no significant differences in the incidence and severity of PONV between the experimental and the control groups.⁽¹³⁾ Therefore, the principal objective of this study was to determine the most effective timing for the application of acupuncture on PC6 in preventing or reducing PONV.

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METHODS

Participants

Patients treated via laparoscopic surgery at the Gunsan Medical Center (GMC), South Korea were recruited through a bulletin board advertisement to participate in acupuncture treatments for the prevention of PONV. Individual patients were eligible to participate in the program if they belonged to category I or II on the Health Grade Classification table of the Association of Anesthesiology of America.⁽¹⁴⁾ Patients with pulmonary, cardiovascular, or cerebral diseases, renal or liver abnormalities, or nausea or who had taken antiemetics 24 h prior to surgery, as well as patients with a past history of PONV or motion sickness were excluded from the study. One hundred seventy-eight patients agreed to participate in this study. Those patients were randomly assigned to either a preoperative EA group (PrEA, 45 cases), a postoperative EA group (PoEA, 53 cases), or a non-acupuncture control group (NC, 51 cases) using simple randomization method with random numbers generated by a computer. All 3 groups received patient-controlled analgesia (fentanyl and ketorolac), and none had received any medication for PONV. The institutional review board of GMC approved this study. All participants and guardians agreed on the objectives and methods of this study and signed informed consent forms prior to participation in this study.

Procedures

Anesthesia was administered to all subjects with 2% lidocaine (40 mg/kg), and propofol (2 mg/kg), and tracheal intubation was facilitated by administering rocuronium (0.6 mg/kg). Additional rocuronium was administered as required. Anesthesia was maintained using 2%–3% sevoflurane (inspired concentration) and 50% nitrous oxide in oxygen. To reverse residual muscle relaxation, a combination of glycopyrrolate (0.008 mg/kg) and pyridostigmine (0.2 mg/kg) was administered and the trachea was extubated. No opioids were administered during the operations.

The PrEA subjects received EA at PC6 at a depth of 1 cm with Deqi during the surgery preparation stage. The PoEA subjects received EA in the recovery room after surgery when they regained their consciousness and were able to communicate with practitioners, which generally took 20 or 30 min.

The acupuncture needles used in this study were stainless steel needles (Dong Bang Needle Company, 0.3 mm × 30 mm, Korea), and electrical stimulation was applied for 15 min at 5 Hz using a Rebirth (SAEIK Medical, Korea) low frequency therapy apparatus. This treatment was to be halted if any adverse events, such as discomfort or dizziness, occurred during treatment; this was made clear to the IRB. However, no adverse events occurred during this study, and thus the data of all the participants were available for analysis.

Outcome Measures

The severity and number of PONV were measured in two ways: a self-reported questionnaire and a confirmation from the anesthetist. The self-reported questionnaire measured the number of times nausea occurred, the number of times vomiting occurred, and the severity of the nausea. The questionnaire categorized the severity of nausea into 4 grades from none (grade 0) to severe (grade 3). The anesthetist, who was blindly to the grouping, then visited the patients on the day of surgery and on the next day (within a 48 h-time-frame), interviewed the patients, and confirmed their charts. This outcome measurement began immediately after surgery when the patients were transferred to their post-operation phase and lasted for 48 h after the surgery.

Statistical Analysis

Data analysis was conducted using SPSS 12.0 for Windows. Data were expressed as $\bar{x} \pm s$; $P < 0.05$ was accepted as having significant differences. In order to detect any common characteristics between the groups, ANOVA analysis was conducted. In order to determine whether or not there were any differences in the incidence of nausea and vomiting between the groups, two proportion comparisons (Z-test) were conducted to compare between the NC and PrEA, NC and PoEA, and the PrEA and PoEA groups, respectively. The Jonckheere-Terpstra test was carried out to evaluate differences in nausea intensity among the 3 groups.

RESULTS

One hundred seventy-eight patients were screened for test eligibility and 149 patients were found to meet the criteria for inclusion in this study. We allocated these patients randomly into 3 groups: PrEA (45 cases), PoEA (53 cases), and NC (51 cases). The study's dropout rates were 4.5% in PrEA group,

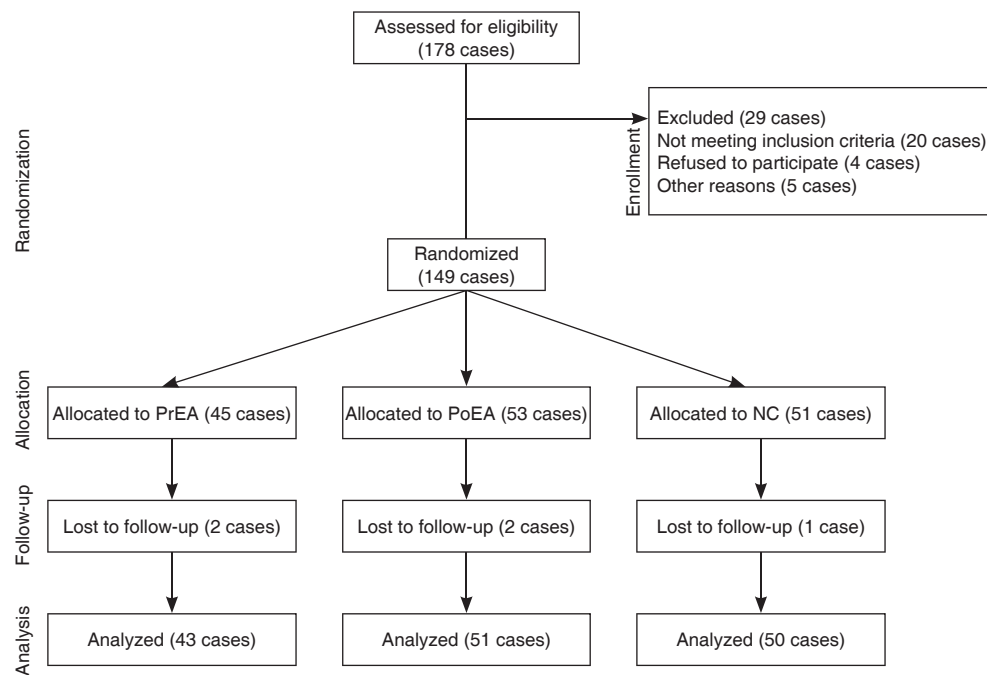


Figure 1. Enrollment of Patients and Completion of the Study

3.9% in PoEA group, and 2% in NC group. Data were available for 43 subjects in the PrEA group, 51 in the PoEA group, and 50 in the NC group (Figure 1).

Comparison of the Baseline Homogeneity among the Three Groups

Table 1 shows the demographic characteristics of subjects in the 3 groups. The 3 groups did not differ significantly in sex, age, height, and weight.

Table 1. Baseline Characteristics of the Three Groups ($\bar{x} \pm s$)

Group	Case	Age (Year)	Height (cm)	Weight (kg)
PrEA	43	47.3 $\pm$ 14.5	161.9 $\pm$ 7.7	60.6 $\pm$ 11.1
PoEA	51	46.3 $\pm$ 13.0	160.2 $\pm$ 7.3	59.6 $\pm$ 10.0
NC	50	44.0 $\pm$ 15.4	161.6 $\pm$ 7.1	60.7 $\pm$ 10.5
<i>P</i> -value		0.520	0.516	0.862

Comparison of the Incidence of Nausea, Vomiting, and Nausea Severity among the Three Groups

The incidence of nausea and vomiting differed significantly among the groups (Table 2). The incidences of nausea and vomiting were significantly lower in the PrEA group than in the NC group ($P < 0.001$ and $P = 0.039$, respectively), and the incidence of vomiting was significantly lower than in the PoEA group ($P = 0.025$). However, no significant differences were determined to exist between the PoEA and the NC groups.

Table 2. The Incidence of Nausea and Vomiting [Case (%)]

Group	Case	Nausea	Vomiting
PrEA	43	8 (18.6)	2 (4.7)
PoEA	51	20 (39.2)	9 (17.6)*
NC	50	27 (54.0)**	8 (16.0)*

Note: * $P < 0.05$, ** $P < 0.01$, compared with the PrEA group

Nausea severity was significantly lower in the PrEA group as compared to the NC and the PoEA groups ($P < 0.05$). Additionally, the severity of nausea was lower in the PoEA group than in the NC group ($P < 0.05$, Table 3).

Table 3. The Severity of Nausea (Case)

Group	Case	0 (None)	1 (Mild)	2 (Moderate)	3 (Severe)
PrEA	43	35	4	2	2
PoEA	51	31	6	7	7
NC	50	23	6	9	12

DISCUSSION

Herein, we have demonstrated the efficacy of PC6 EA in reducing the incidence of PONV. We also demonstrated that the appropriate PC6 EA should be carried out prior to noxious events that might induce PONV, such as surgery or anesthesia.

In this study, the incidence of nausea arising in

the 48 h after surgery were 18.6% in the PrEA group, 39.2% in the PoEA group, and 54% in the NC group. Considering that the incidence of PONV in general cases was 30%–79%⁽¹⁵⁾ and in cases of Fentanyl use was 80%,⁽¹⁶⁾ the incidence of the control group in this study was generally consistent with previous results.

In the PrEA group, the incidences of nausea and vomiting were significantly lower than in the NC group, which supports the assumption that the acupuncture application time is important for enhancing the efficacy of acupuncture, in this study, especially before the noxious events. In previous studies, PC6 EA or acustimulation have been variously shown to be effective and non-effective relative to the controls. There has been some controversy as to the reason for these discrepancies in the effects of PC6 stimulation; some have proposed that this is attributable to differences in the type of surgery being performed.⁽¹⁷⁻²⁰⁾ Recently, Lee A, et al⁽¹²⁾ conducted a systematic review of studies involving PC6 stimulation, and reported that it reduced the risk of nausea, but that further studies would be necessary to adequately assess its effects, depending on when the acupuncture was applied. Dundee, et al⁽²¹⁾ also reported that the reason that studies into the effects of PC6 yielded different results was attributable to different times of PC6 stimulation, and that PC6 stimulation was only effective when applied before the factors causing nausea arose.

The anti-emetic effects of acupuncture may be attributable to changes in the activity of neurochemicals, including endorphin, serotonin, and norepinephrine, in the central nervous system. When an acupuncture point is stimulated, a low frequency electrical stimulus is generated, which activates the A- β and A- δ fibers, thus releasing endorphins into the central nervous system, which ultimately strengthens the intrinsic anti-vomiting pathway. Reviewing the manner in which the central serotonin and norepinephrine fibers are activated, it appears that a change occurs in the serotonin transmission pathway, resulting in anti-vomiting effects.⁽¹⁷⁾ The neurochemicals released by acupuncture point stimulation desensitize the vomiting center in the brain, resulting in the prevention of nausea and vomiting. It is also suspected that once the vomiting center in the brain is sensitized, it is difficult to desensitize or overcome it.

This study also shows that the PoEA group evidenced significantly lower response in terms of nausea severity than the control group, but no significant response in the incidence of nausea and vomiting relative to the control group. These results may attributable to the fact that the PoEA group already experienced strong PONV-inducing events: namely, anesthesia, surgery and opioid use.

The PC6, which is located on the pericardium meridian, has mind-relaxing, calming, and digestive effects, and is known to be effective against palpitation, chest pain, and gastritis. It meets the Triple Energizer meridian, and is applied to a variety of chest and lower stomach illnesses. This point, which is the most frequently used acupuncture point in the treatment of PONV, can also be stimulated by the patient himself.⁽²²⁾

This study had a few limitations. First, the number of participants was not sufficiently large for our findings to be confirmed. Second, we were unable to achieve blinding in the study design, since when we defined the experimental and control groups, we did not apply a placebo treatment, such as sham acupuncture, to the control group. Third, even though we conducted our study on laparoscopic surgery patients who had undergone PCA treatment, the surgeries received by each patient were not identical. This study employed the PC6 alone for treatment; however, under clinical conditions, when nausea and vomiting symptoms arise, the PC6 – Gongsun (SP4) and PC6 – Zusanli (ST36) points are used together. Clearly, more study will be required to evaluate the efficacy of grouped-acupuncture stimulation.

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